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Morphology-Controllable Synthesis of Microporous Prussian Blue Analogue $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ Microstructures

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Microporous prussian blue analogue $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ microspheres and micropolyhedrons self-assembled by nanoparticles were synthesized under ultrasonic conditions using poly(vinylpyrrolidone) (PVP) as a surfactant. Obvious self-assembly behavior was observed when the microparticles dispersed in alcohol were dropped on a copper grid or a glass substrate without any further treatment. N_2 adsorption properties confirmed the existence of micropores in both $Zn_3[Co(CN)_6]_2$ microspheres and micropolyhedrons. The concentration of reactants and the molar ratio of the reactant to the PVP as well as ultrasonic conditions have important effects on the shape and size of the product. A possible mechanism is proposed.

1. Introduction

Recently, growing attention has been paid to Prussian blue (PB) and its analogues for their applications in hydrogen storage,[1] molecular magnetic,[2,3] optics,[4] and so on.[5] Many studies related to them have been reported. For example, Mann's group reported the synthesis of Prussian blue nanoparticles by photoreduction of $[Fe(C_2O_4)_3]^{3-}$ in the presence of $Fe(CN)_6{}^{3-}$ ions;[6] Hashimoto's group has reported photomagnetic behavior based on PB analogues such as photoinduced magnetization, photodemagnetization, and magnetic pole inversion;[7a] Taguchi recently reported the photocontrolled magnetization of CdS-modified Prussian blue because of the crucial photoinduced electron transfer from CdS to PB;[7b] Uemura reported PB nanoparticles protected by poly(vinylpyrrolidone) (PVP) and poly(diallyldimethylammonium chloride) (PDDA), which exhibited a size-dependent property;[8] our group has done some work on the synthesis of Prussian blue and its analogue $Co_3[Co(CN)_6]_2$ from a single-source precursor;[9] Long and Kaye reported the hydrogen storage properties of hydrated prussian blue analogues $M_3[Co(CN)_6]_2$ (M = Mn, Fe, Co, Ni, Cu, and Zn),[1] in which it has been addressed that it is the interactions between the hydrogen and bridging cyanides as well as the unique structure of the dehydrated prussian blue that lead to the excellent gas adsorption (desorption) properties. Whereas in microporous metal organic framework materials, such as $Zn_4(OBDC)_3$ (BDC = 1,4-benzenedicarboxylate), the primary reason for gas adsorption is van der Waals interaction with the framework.[10-15] So if these two favorable facets can be combined together in one compound, the adsorption effect may be much better. Our as-synthesized $Zn_3[Co(CN)_6]_2$ samples here, uniform microspheres or micropolyhedrons, which are composed of nanoparticles (see SEM pictures) and have a microporous structure (verified by the N_2 adsorption experiment), are characterized by the two features mentioned above.

"Self-assembly"[16-20] becomes more and more popular nowadays in scientific research area, and a lot of research papers emerge swiftly in a short period. There is an increasing need for the fabrication of different dimensional structures of magnetic nanoparticles through self-assembly using nanoscale building blocks.[21-23] The integrity of the building blocks based on self-assembly might provide an intriguing strategy for designing frameworks with desirable shapes and sizes.[24] Yu and Yam reported that Ag nanoparticles were able to self-assemble into two-dimensional arrays on a solid substrate, in which the surfactant prevented the particles from aggregating and thus the uniformity of the nanoparticles made it possible to manipulate the cubes into a close-packed and ordered array caused by van der

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Table 1. Summary of the Main Results on the Products Obtained under Different Preparation Conditions

sample	$ZnAc_2$ (mmol)	PVP (mmol)	$K_3[Co(CN)_6]_2$ (mmol)	morphology
1	0.6	10	0.4	sphere
2	0.3	10	0.2	sphere
3	0.15	10	0.1	spheres and polyhedrons
4	0.075	10	0.05	polyhedron
5	0.06	10	0.04	polyhedron

Waals forces.[25] Meanwhile, the shape of the nanoparticles can induce the texture of the self-assembly superlattice arrays, which means that the structural alignment of each particle can be achieved if the morphology of the nanoparticles is controlled.[26] Such structure can lead to important properties such as anisotropic magnetism.[27a] Generally, good control over size and shape can result in better uniformity of nanoparticles and further self-assembly.

In the present work, microporous $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ microspheres and micropolyhedrons were synthesized by using PVP as surfactant under ultrasonic conditions at room temperature. In fact, these microstructures are composed of nanoparticles with a size of about 50 nm. What has excited us most is that evident self-assembly behavior was observed when the samples dispersed in alcohol were dropped on a copper grid or a glass substrate without any further treatment. This method is also applicable to other analogues such as $Co_3[Co(CN)_6]_2 \cdot xH_2O$, $Mn_3[Co(CN)_6]_2 \cdot xH_2O$, $Ni_3[Co(CN)_6]_2 \cdot xH_2O$, and $Cu_3[Co(CN)_6]_2 \cdot xH_2O$. To the best of our knowledge, it is the first time for synthesis of these microporous PB microstructures with superior gas adsorption properties.

2. Experimental Section

All chemicals were analytical grade and used without purification. The typical synthetic experiments were as follows.

Synthesis of $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ Microspheres. Solution A: 0.6 mmol of $Zn(CH_3COO)_2 \cdot 2H_2O$ was dissolved in 20 mL of distilled water, and then 10 mmol of PVP was added under agitated stirring to get an absolute transparent solution. Solution B: 0.4 mmol of $K_3[Co(CN)_6]_2$ was dissolved in 20 mL of distilled water with stirring to get a homogeneous solution. Solution B was added into solution A slowly and regularly under stirring and ultrasonic conditions. Notice that the whole reaction process was kept at a constant temperature of 20 °C by water circulation. Afterward, the reaction was kept under only ultrasonic conditions. After 1 h, the reaction was aged under room temperature without any interruption for 24 h. The resulting white precipitation was filtered and washed several times with distilled water and absolute ethanol and finally dried naturally.

Synthesis of $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ Micropolyhedrons. For the synthesis of $Zn_3[Co(CN)_6]_2$ micropolyhedrons, only the molars of the $Zn(CH_3COO)_2 \cdot 2H_2O$ and $K_3[Co(CN)_6]_2$ were decreased to 0.06 and 0.04 mmol, respectively, and other procedures were same as that of $Zn_3[Co(CN)_6]_2$ microspheres.

Characterization. X-ray powder diffraction (XRD) was performed on a Japan Rigaku D/max γA X-ray diffractometer with graphite-monochromatized Cu Kα radiation (λ = 1.5406 nm). A JEOL JEM-2010F transmission electron microscope operating at 200 kv accelerating voltage was used for transmission electron microscopy (TEM) analysis. Scanning electron microscopy (SEM) was performed with an Amray 1910FE microscope. C, H, and N elemental analyses were carried out on an Elementar Analysensysteme GmbH VarioEL instrument. N_2 adsorption−desorption characterization was carried out on a Nova 1000 analyzer at liquid nitrogen temperature (77 K). Superconducting quantum interference device (SQUID) magnetic characterization of the microcapsules was performed on a Quantum Design magnetometer (MPMS XL 7). Zero-field and field-cooled runs were performed between 2 and 300 K in a static magnetic field of 100 Oe. For the magnetization measurements, samples were loaded into a gelatin capsule and the measurements were corrected for the diamagnetic contribution of the sample holder.

3. Results and Discussion

3.1. X-ray Powder Diffraction (XRD) of the Products. Samples obtained under different reaction conditions are listed in Table 1.

Figure 1A shows typical XRD patterns of as-synthesized microsphere and micropolyhedron samples. All the diffraction peaks for the microsphere sample can be indexed cubic $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ phase with lattice constant of a = 10.26 Å (PCPDF No. 32-1468). The sharp peaks indicate that the as-obtained sample has a good crystallinity. However, for the micropolyhedron sample, a noticeable difference is observed in the 2θ (12−70°) range with all peaks being shifted by more than 0.08° toward smaller angles as compared with the microsphere sample. This should correspond to an increase in the crystallographic cell volume,[27b] which may be attributed to different water of crystallization contents in the samples. In addition, it was recently reported that $Zn_3[Co(CN)_6]_2$ is dimorphic and is cubic phase ($Fm\bar{3}m$) when hydrated, while the anhydrous phase becomes rhombohedral phase ($R\bar{3}c$). So, to verify this kind of phase transformation, the $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ spheres and polyhedrons were completely dehydrated in vacuum at 120 °C for 48 h, and then XRD measurement of the resulting samples was carried out quickly. As shown in Figure 1B, the XRD patterns for both dehydrated samples are in good agreement with the literature without any peak shift.[28] This result further confirms that it is different water of crystallization contents that result in the above peak shift.

3.2. Morphology Characterizaition and Effect of the Reaction Conditions on the Morphology. When the concentration of $Zn(NO_3)_2 \cdot 6H_2O$ was 0.015 M and the molar ratio of PVP to $K_3[Co(CN)_6]$ was 25, uniform spheres with a diameter of 1.0 μm were obtained, whereas when the molar concentration of the $Zn(NO_3)_2 \cdot 6H_2O$ was decreased to 0.0015 M and the molar ratio of PVP to $K_3[Co(CN)_6]$ was increased to 250, fine micropolyhedrons instead of microspheres self-assembled in an orderly manner on glass substrate were observed. Figure 2 displays typical SEM images of $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ microspheres and micropolyhedrons at different magnifications. From the low magnification SEM images (Figure 2a,c), it can be seen that the sample is composed of quite uniform microspheres (a) and micropolyhedrons (c). Furthermore, microspheres and micropolyhedrons have a narrow size distribution and their surfaces are quite rough as we can see obviously from the high magnification SEM images

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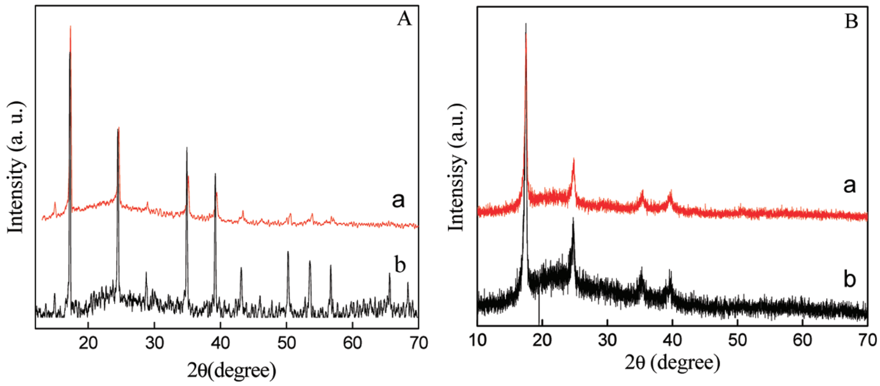


Figure 1. (A) Typical XRD patterns of (a) $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ microspheres obtained with the concentration of the $Zn(NO_3)_2 \cdot 6H_2O$ of 0.015 M and the molar ratio of PVP to $K_3[Co(CN)_6]_2$ at 25 and (b) $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ micropolyhedrons obtained with the concentration of the $Zn(NO_3)_2 \cdot 6H_2O$ of 0.0015 M and the molar ratio of PVP to $K_3[Co(CN)_6]_2$ at 250. (B) Typical XRD patterns of $Zn_3[Co(CN)_6]_2$ microspheres (a) and micropolyhedrons (b) after dehydrating at 120 °C for 48 h in vacuum.

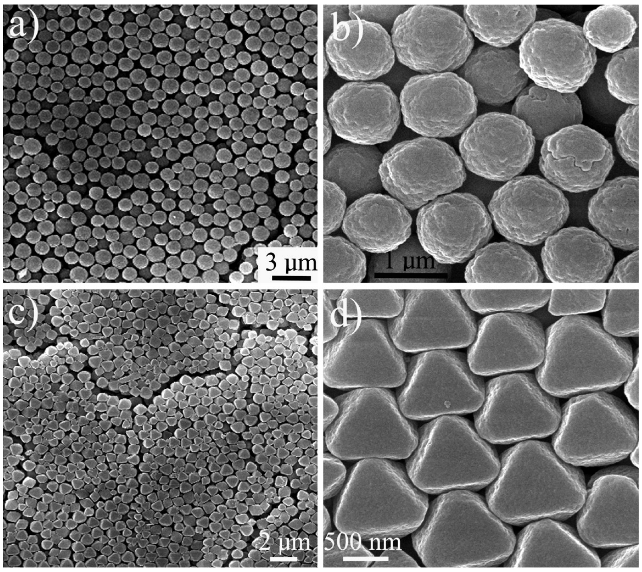


Figure 2. SEM images of $Zn_3[Co(CN)_6]_2$ microspheres obtained with the concentration of $Zn(NO_3)_2 \cdot 6H_2O$ of 0.015 M and the molar ratio of PVP to $K_3[Co(CN)_6]$ at 25 (a) at low-magnification and (b) at high-magnification. SEM images of $Zn_3[Co(CN)_6]_2$ micropolyhedrons obtained with the concentration of $Zn(NO_3)_2 \cdot 6H_2O$ of 0.0015 M and the molar ratio of PVP to $K_3[Co(CN)_6]$ at 250 (c) at low magnification and (d) at high magnification.

(Figure 2b,d). TEM images (Figure 3a,b) also confirm the narrow size distribution of these microspheres and micropolyhedrons. Further research proved the fact that both microspheres and micropolyhedrons are formed by self-organizing of nanoparticles with a size of around 50 nm, which could be confirmed by fragments (Figure 3c,d) obtained after they were treated under ultrasonic conditions for at least 30 min. In addition, we also tracked the TEM images of the intermediate products during the reaction process by taking a sample from the reaction system at different ultrasonic times. Figure 3e,f shows the TEM patterns of the microsphere sample obtained with ultrasonic times of 10 and 15 min, respectively. It can be clearly seen that the product consists of small nanoparticles of about 50 nm in diameter with an ultrasonic time of 10 min (Figure 3e). When the time is increased to 15 min, the particles start to aggregate to form fused microspheres (Figure 3f). These results further confirm the above fact.

A series of compared experiments have been carried out to investigate the influence of all possible conditions on the

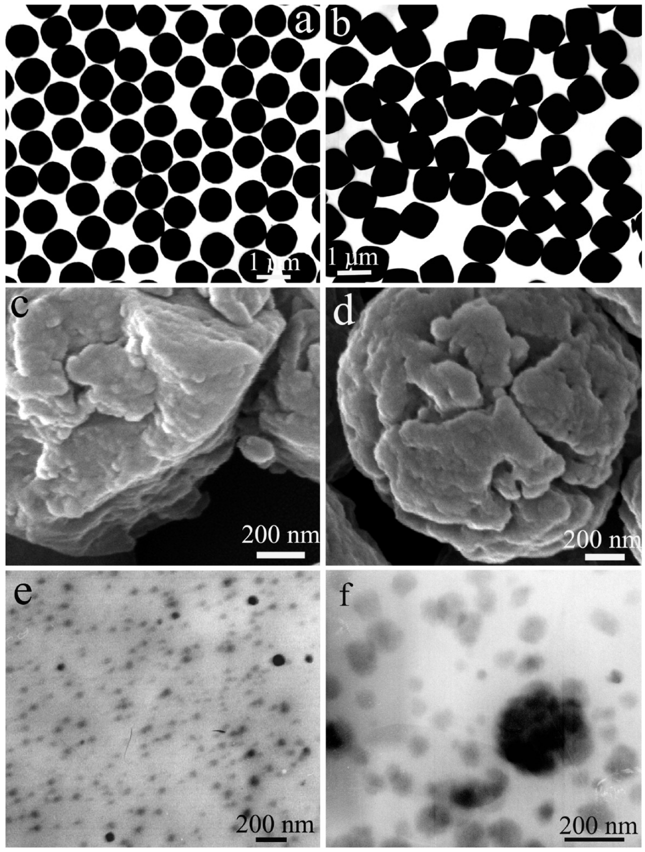


Figure 3. TEM images of $Zn_3[Co(CN)_6]_2$ microspheres obtained with the concentration of the $Zn(NO_3)_2 \cdot 6H_2O$ of 0.015 M and the molar ratio of PVP to $K_3[Co(CN)_6]$ at 25 (a) and $Zn_3[Co(CN)_6]_2$ micropolyhedrons obtained with the concentration of the $Zn(NO_3)_2 \cdot 6H_2O$ of 0.0015 M and the molar ratio of PVP to $K_3[Co(CN)_6]$ at 250 (b). SEM images of same samples after ultrasonic treatment for 30 min (c) polyhedrons and (d) microspheres. TEM images of same microsphere samples after ultrasonic treatment for 10 min (e) and 15 min (f).

morphology and size of the product. When the concentrations of reactants ($K_3[Co(CN)_6]$ and $ZnAc_2$) both decreased as listed in Table 1, the morphology of the product changed from sphere to polyhedron as shown in Figure 3a,b. In addition, PVP also plays an important role on the morphology of the product. If no PVP was used but with other conditions being the same as those for the above microspheres and micropolyhedrons, irregular particle

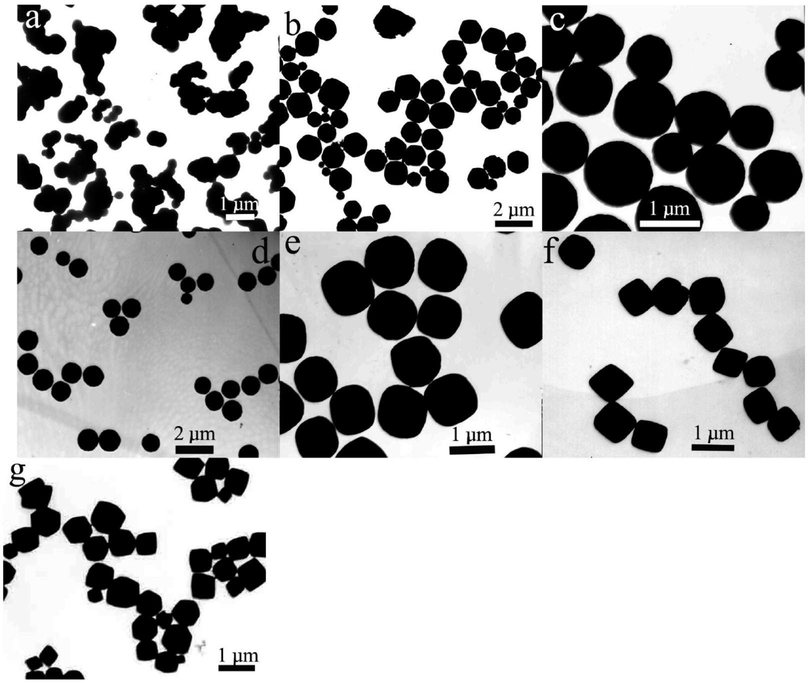
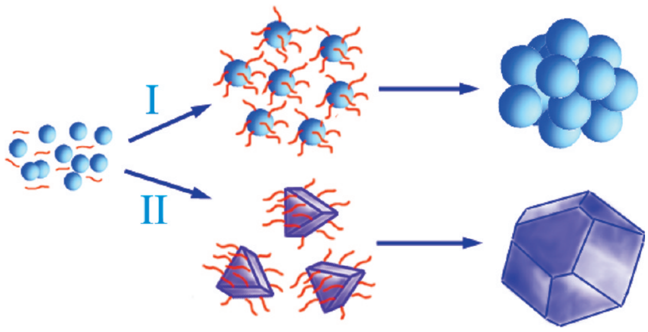


Figure 4. TEM images of samples obtained without the presence of PVP, but with other conditions being the same as those for sample 1 (a) and sample 4 (b) listed in Table 1. TEM images of samples obtained with the concentration of $Zn(CH_3COO)_2 \cdot 2H_2O$ at 0.015 M and the concentration of PVP at 0.0025 M (c) and 0.375 M (d). TEM images of samples obtained with the concentration of $Zn(CH_3COO)_2 \cdot 2H_2O$ at 0.001875 M and the concentration of PVP at 0.0025 M (e) and 0.5 M (f). (g) TEM image of sample obtained without using ultrasonic condition, but with other conditions being same as those for sample 1 listed in Table 1.

agglomerates and nonuniform polyhedrons were formed (as shown in Figure 4a,b, respectively), which illustrated that PVP contributed greatly to forming spherically aggregated particles or uniform polyhedrons and effectively preventing the small nanoparticles from random agglomeration. The concentration of PVP was changed according to two different ways to investigate its effect on the morphology of the product. First, keeping the concentration of $Zn(CH_3COO)_2 \cdot 2H_2O$ at 0.015 M unchanged and varying the concentration of PVP from 0.0025 to 0.375 M, the morphology changed from irregular aggregation of nanoparticles to uniform spheres and the diameter altered from 1 to 0.8 μm (Figure 4c,d). Second, keeping the concentration of $Zn(CH_3COO)_2 \cdot 2H_2O$ at 0.001875 M and varying the concentration of PVP from 0.0025 to 0.5 M, the morphology changed from spherelike polyhedrons to perfect octahedrons without obvious size change (Figure 4e,f). In addition, the ultrasonic condition is another important factor influencing the uniformity and shape of the product. Only when its intensity reached a certain degree could uniform microparticles be achieved, but the particles became bigger. On the contrary, when the experiment proceeded without ultrasonic conditions, only irregular particles were obtained as shown in Figure 4g. So it can be concluded that the ultrasonic conditions, on the one hand, contributed greatly to the uniformity and regularity of product, but on the other hand they accelerated the chemical reaction, easily leading to the formation of the bigger size particles.

3.3. Formation Mechanism of $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ Microspheres and Micropolyhedrons. It can be seen from the above experimental results that the molar ratio of PVP to $Zn(CH_3COO)_2 \cdot 2H_2O$ plays a crucial role for the formation of microparticles with different shapes. The possible formation process of microspheres and micropolyhedrons can be proposed as illustrated in Scheme 1. When the lower molar ratio of PVP to $Zn(CH_3COO)_2 \cdot 2H_2O$ was used, $Zn_3[Co(CN)_6]_2$ crystal nuclei were not completely capped by PVP, they grew almost equally

Scheme 1. Schematic Illustration of the Growth Mechanism for the Formation of $Zn_3[Co(CN)_6]_2$ Microspheres and Micropolyhedrons



in all directions with little limit of crystal epitaxy, and they finally formed and agglomerated into spherically shaped particles to realize the lowest surface energy (Scheme 1-I). However, with increasing molar ratio of PVP to $Zn(CH_3COO)_2 \cdot 2H_2O$, the $Zn_3[Co(CN)_6]_2$ crystal nuclei were fully capped by PVP, and PVP was selectively adsorbed on some specific crystal surfaces of the $Zn_3[Co(CN)_6]_2$ crystal nuclei,[29−31] which on the one hand could effectively lower the energy of particles and on the other hand induce particle epitaxy and assembly into micropolyhedrons (Scheme 1-II). Without PVP, only irregular aggregations of nanoparticles were obtained, which further confirmed the capping agent function of PVP in our work. The literature suggested that the energy difference of crystal facets ⟨111⟩ to ⟨100⟩ could lead to the formation of polyhedrons.[32] So we speculated that

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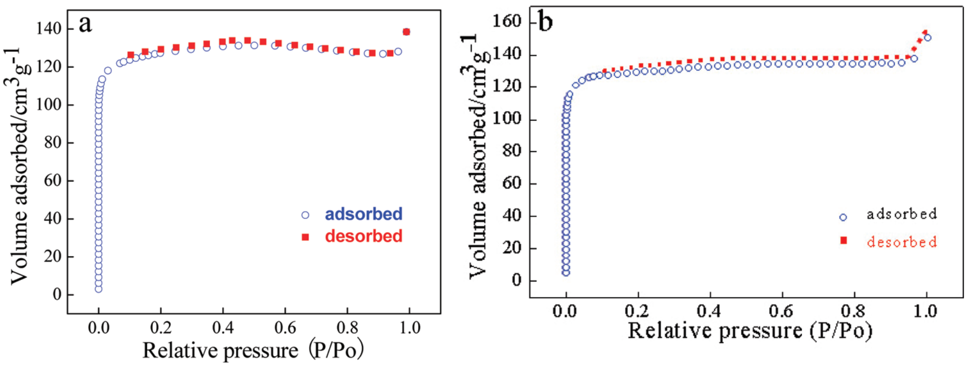

Figure 5. Nitrogen adsorption−desorption isotherms of $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ microspheres (a) and micropolyhedrons (b).

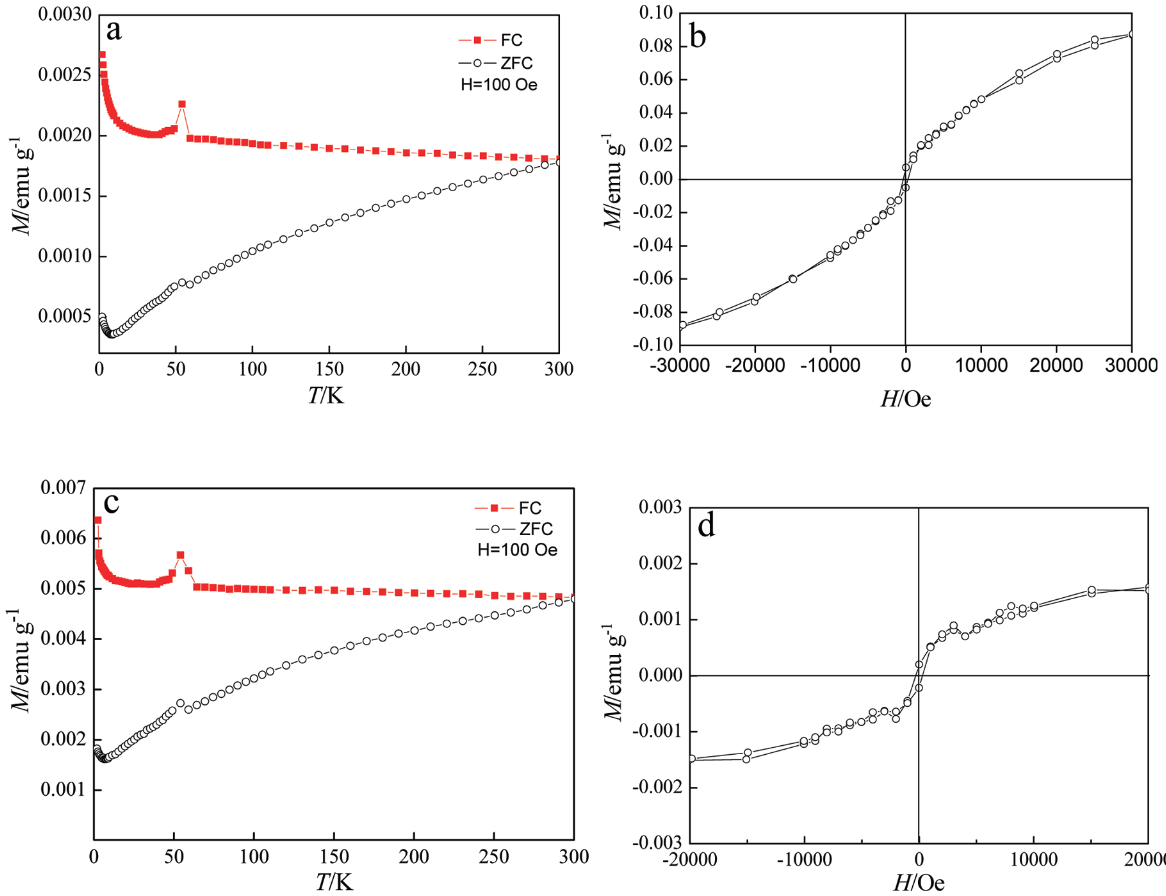

Figure 6. FC and ZFC magnetism versus temperature curves for $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ microspheres (a) and micropolyhedrons (b); magnetization versus applied magnetic field hysteresis loop at 2 K for $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ microspheres (c) and micropolyhedrons (d).

both these two reasons contribute to the production of regular polyhedrons.

To testify to the existence of PVP in our samples in a quantifiable way and accordingly explain the capping function of PVP, C, H, and N elemental analyses were carried out. For the microsphere sample, the C, H, and N elemental weight percents were 27.87%, 3.811%, and 18.90%, respectively; for the polyhedron sample, the corresponding data were 18.83%, 3.28%, and 19.35%, all of which were higher than the theoretically calculated weight percents (C, 15.76%; H, 3.53%; and N, 18.38%). This result shows that PVP existed in both spheres and polyhedrons. Furthermore, it can be also found from the elemental analyses that the PVP weight percent in the microsphere sample is higher than that in the micropolyhedron sample. It is speculated that, in the reaction process of spheres, the reaction was so fast that the capped PVP molecules cannot desorb from the crystals and they are wrapped in the spheres, resulting in the high content of PVP in the final product, whereas the reaction of polyhedrons was slow, so the PVP content is relatively low.

3.4. N_2 Adsorption−Desorption Properties of $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ Microspheres and Micropolyhedrons. In order to examine the porous structure of these microspheres and micropolyhedrons, the N_2 adsorption properties of the microsphere and micropolyhedron samples were measured. Figure 5 shows the N_2 adsorption/desorption isotherms of $Zn_3[Co(CN)_6]_2$ microspheres (a) and micropolyhedrons (b). Both of them were measured at 77 K. Before measurement, the samples were heated at 95 °C for 55 h under vacuum to dehydrate completely as previously described in the literature.[1] Both the isotherms are identified as type I,[33−36] which is characteristic of the microporous materials. The Brunauer−Emmett−Teller (BET) specific surface areas of spheres and polyhedrons were found to be 428.20 and 441.36 m^2 g^{-1}, respectively, and the total pore volumes were 0.2157 and 0.2395 cm^3 g^{-1}, respectively. The relatively high BET surface area and large total pore volume in our case have potential applications in gas absorption.

3.5. Magnetic Properties of $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ Microspheres and Micropolyhedrons. The detailed magnetic properties of the $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ microspheres and micropolyhedrons have been studied by using a superconducting quantum interference device (SQUID) magnetometer. We have recorded temperature-dependent zero-field-cooled (ZFC) magnetization, field-cooled (FC) magnetization, and field-dependent magnetization measurements for both samples. As shown in Figure 6, it can be clearly seen that microspheres and micropolyhedrons exhibit almost completely same magnetic properties. The reason may be because microspheres and micropolyhedrons both are constructed from the same building blocks (nanoparticles of 50 nm). Therefore, in the following, we will use microspheres as an example to discuss magnetic properties. As we know, ZFC and FC measurements have been proved to be a useful way of determining the so-called blocking temperature (T_B) of magnetic nanomaterials. As shown in Figure 6a, the ZFC curve in the range of 0−300 K was much lower than the FC curve, and the FC data deviate from the ZFC data well above 300 K, suggesting ferromagnetic characteristics of as-prepared samples at room temperature. The origin of the high T_B features may originate from the high anisotropic energy of the as-prepared samples.[37] Because the as-synthesized samples in our case consist of nanoparticles, we believe that the anisotropy of the nanoparticles must be high. The magnetization versus applied field for both samples is shown in Figure 6b,d. The magnetic hysteresis could be clearly observed at 2 K, although it is much weaker, indicating their weak ferromagnetic properties at 2 K.

4. Conclusions

We have synthesized uniform Prussian blue analogue $Zn_3[Co(CN)_6]_2 \cdot xH_2O$ microspheres and micropolyhedrons assembled by nanoparticles through a simple precipitation method under ultrasonic conditions using PVP as a capping agent. Microparticles could self-assemble regularly on a glass substrate or a copper grid after slow evaporation of the ethanol suspension of the sample. Both spheres and polyhedrons are aggregations of nanoparticles. N_2 adsorption measurements proved that both microspheres and micropolyhedrons exhibit a microporous structure, which may have very important application in future gas adsorption.

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